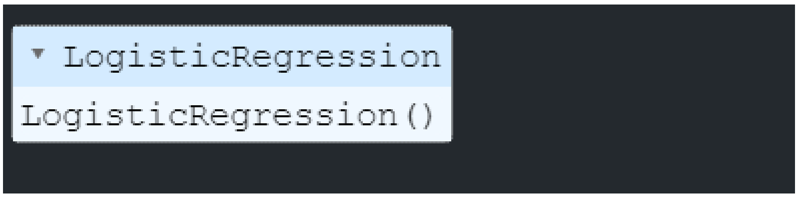
```python
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import confusion_matrix, accuracy_score, precision_score, recall_score

df = pd.read_csv(r"C:\Users\ACER\Downloads\archive (1)\Social_Network_Ads.csv")

df.head()
```

	User ID	Gender	Age	EstimatedSalary	Purchased
0	15624510	Male	19	19000	0
1	15810944	Male	35	20000	0
2	15668575	Female	26	43000	0
3	15603246	Female	27	57000	0
4	15804002	Male	19	76000	0

```python
X = df[['Age', 'EstimatedSalary']]
Y = df['Purchased']
X_train, X_test, Y_train, Y_test = train_test_split(X, Y, test_size=0.2, random_state=42)
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

```
model = LogisticRegression()

model.fit(X_train, Y_train)
```




```
Y_pred = model.predict(X_test)

cm = confusion_matrix(Y_test, Y_pred)

TN, FP, FN, TP = cm.ravel()


print(f"True Positives (TP): {TP}")

print(f"False Positives (FP): {FP}")

print(f"True Negatives (TN): {TN}")

print(f"False Negatives (FN): {FN}")
```

True Positives (TP): 19

False Positives (FP): 2

True Negatives (TN): 50

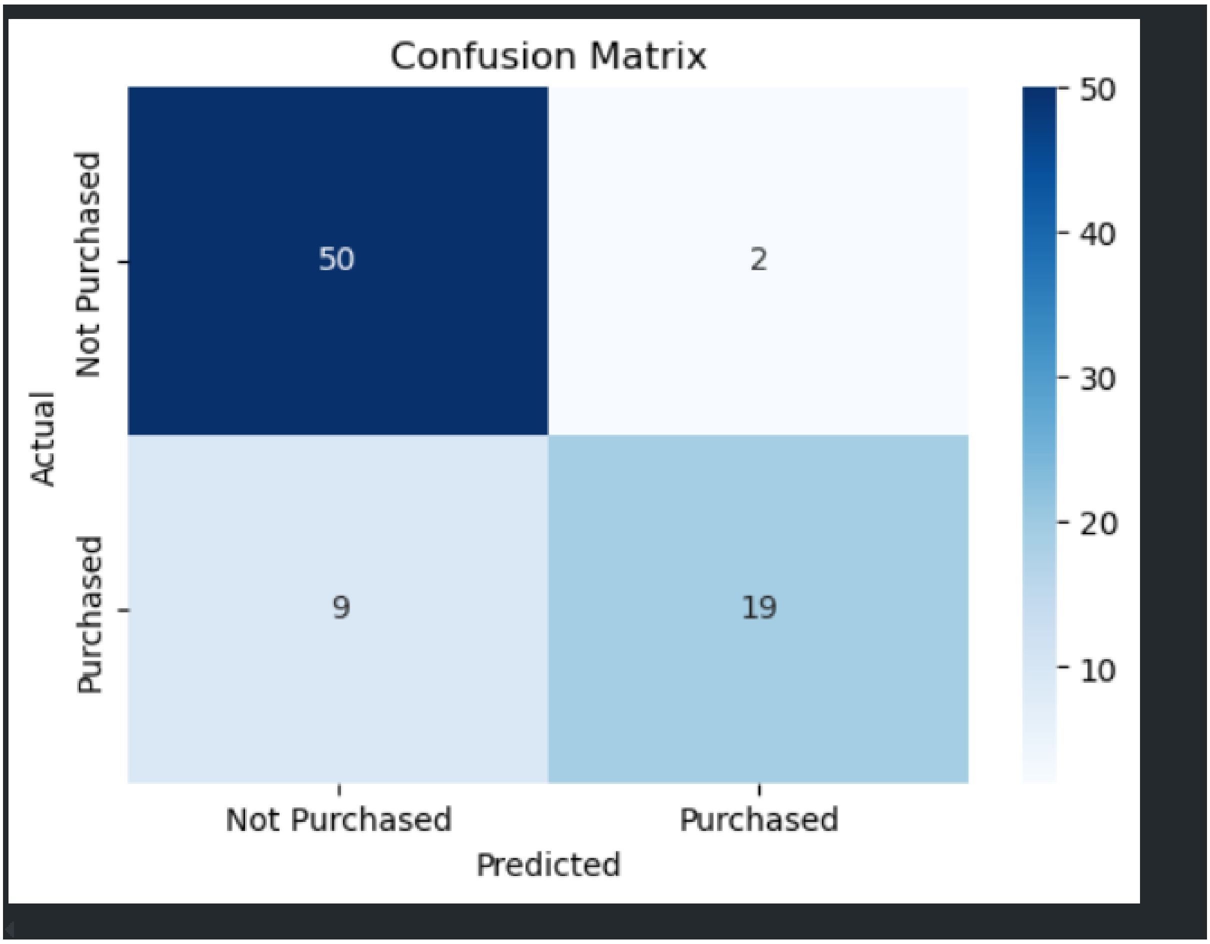
False Negatives (FN): 9

```
accuracy = accuracy_score(Y_test, Y_pred)

error_rate = 1 – accuracy

precision = precision_score(Y_test, Y_pred)

recall = recall_score(Y_test, Y_pred)

from sklearn.metrics import f1_score

f1 = f1_score(Y_test, Y_pred)
```

```
print(f"Accuracy: {accuracy:.2f}")
print(f"Error Rate: {error_rate:.2f}")
print(f"Precision: {precision:.2f}")
print(f"Recall: {recall:.2f}")
print(f"F1 Score: {f1:.2f}")
```

```
Accuracy: 0.86
Error Rate: 0.14
Precision: 0.90
Recall: 0.68
F1 Score: 0.78
```

```
plt.figure(figsize=(6, 4))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues', xticklabels=['Not Purchased', 'Purchased'], yticklabels=['Not Purchased', 'Purchased'])
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")
plt.show()
```


Confusion Matrix
Not Purchased
Purchased
Actual
Not Purchased
Purchased
Predicted
50
2
9
19
50
40
30
20
10